

Nirajan Tripathi

GIS Analyst · Geospatial Data Scientist · Remote Sensing

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SUMMARY

GIS and geospatial data professional with 4+ years across spatial analysis, remote sensing, and field surveying, plus graduate research in hydroclimatology. I build reproducible, end-to-end geospatial workflows across Google Earth Engine, PostGIS spatial SQL, Python/R, and web mapping, each shipping with documented code and verifiable results. Seeking GIS Analyst, GIS Developer, or geospatial data science roles in environment, water resources, energy, and planning.

TECHNICAL SKILLS

GIS & Mapping: ArcGIS Pro, QGIS, ArcGIS Online, ModelBuilder, Field Maps, Survey123

Remote Sensing: Google Earth Engine, ERDAS IMAGINE, Landsat / Sentinel, NDVI / LST, supervised classification, hyperspectral

Programming & Data: Python (ArcPy, Pandas, NumPy, Rasterio), R (Tidyverse, sf), SQL, PostGIS spatial SQL, Git / GitHub

Web Mapping: Leaflet, Chart.js, GeoJSON, HTML / CSS / JavaScript

Surveying & Field: GNSS / DGPS, UAV / drone mapping, Pix4D, AutoCAD

SELECTED PROJECTS

Austin Urban Heat Island Story Map

Earth Engine · PostGIS · R · Leaflet

- Measured summer land surface temperature across all 65 Austin neighborhoods from Landsat 8/9 thermal imagery in Google Earth Engine; found strong correlations with tree canopy ($r = -0.87$) and impervious surface ($r = +0.83$).
- Joined U.S. Census tract income to neighborhoods with an **area-weighted PostGIS spatial join** (coordinate reprojection, geometry validation), and published the analysis as an interactive scrollytelling web map.

Rooftop Solar Across Data Environments

ArcGIS Pro · LiDAR · Python

- Ran an identical rooftop-solar suitability method on 1 m airborne LiDAR (Austin) and a 30 m open DSM (Kathmandu) to **quantify how elevation-data resolution systematically biases** solar estimates.

Watershed Land Cover Simulation (Markov-CA)

ArcGIS Pro · Python · NLCD

- Built a Markov chain–Cellular Automata model that learns 2010–2020 land-cover transitions and forecasts urban growth to 2030, validated at a **Figure of Merit of 0.85**.

Transit Dependency Likelihood Index – Portland

ArcPy · Network Analyst

- Developed a custom ArcPy tool combining demographic need with transit service areas (z-scores, VIF screening) to identify high-need neighborhoods underserved by bus stops.

NDVI, Land Surface Temperature & Land Cover

Landsat · R

- Quantified vegetation cooling across five land-cover types in the Texas Hill Country with a reproducible R workflow (Kruskal–Wallis, ANCOVA).

EXPERIENCE

Graduate Research Assistant · Texas State University, San Marcos, TX

Jan 2025 – Present

- Analyze hydroclimatic variability and trends using Python and R (statistical and time-series methods) to support climate-risk and water-resources assessment.
- Build reproducible data-processing workflows for multi-source national climate and hydrologic datasets, improving consistency and repeatability across analyses.

Geomatics Engineer · Pokhara Metropolitan City, Gandaki, Nepal

Nov 2023 – Jun 2024

- Conducted GNSS and UAV surveys for river, terrain, and surface-water analysis supporting flood-risk assessment.
- Mapped multi-temporal water-extent change in Fewa Lake and set high-flood-level benchmarks to support urban flood-risk management.

GIS Analyst · Earth Eco Consultancy, Gandaki, Nepal

Jun 2021 – Oct 2023

- Performed multi-criteria suitability analysis (MCDA) for land-use planning and built a wildfire-risk model for Banke National Park to support fireline planning.
- Delivered DGPS cadastral / land-pooling surveys and municipal-scale thematic mapping for planning and stakeholder reporting.

EDUCATION & AWARDS

M.S. Geography, Texas State University (GPA 3.84 / 4.0)

Aug 2024 – Present

Awards: The Graduate College Scholarship (2026); Joe & Jerry Moore Scholarship (2025); TSUS Graduate Endowed Fellowship (2024 & 2025); TXST Graduate Merit Fellowship (2024).

B.S. Geomatics Engineering, Tribhuvan University, Nepal

Aug 2016 – Apr 2021